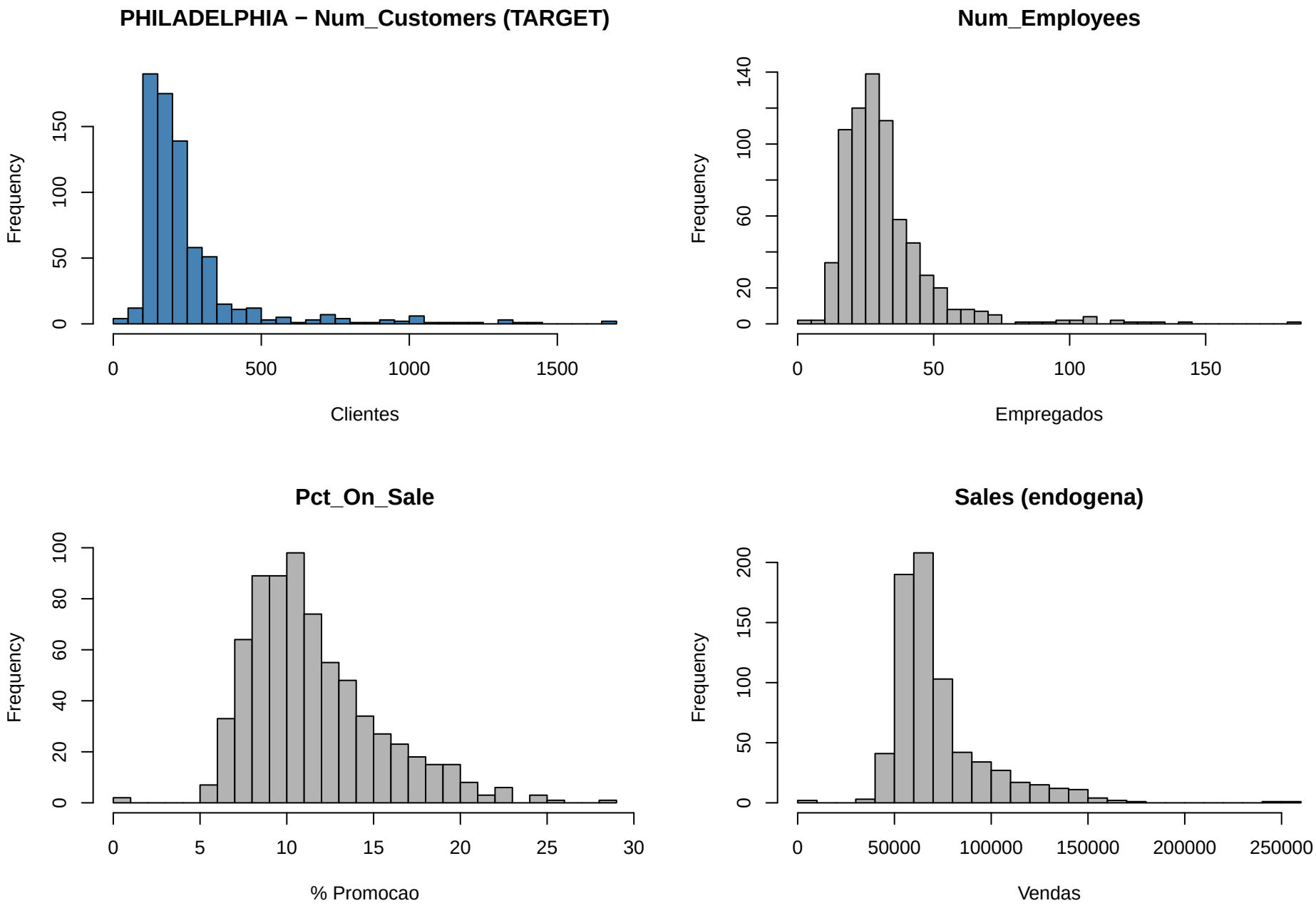
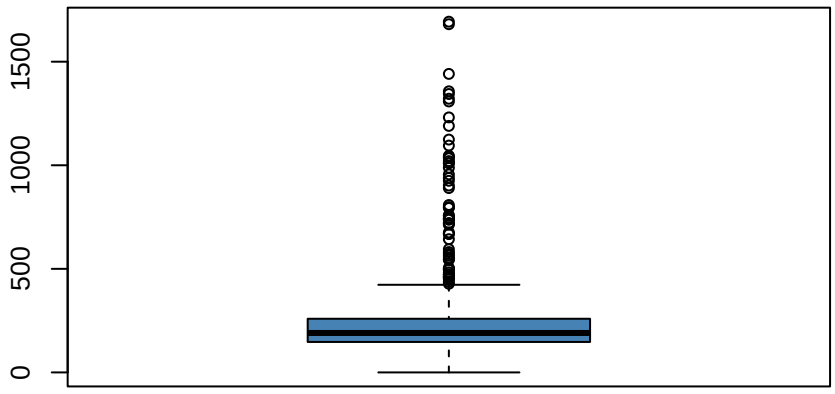
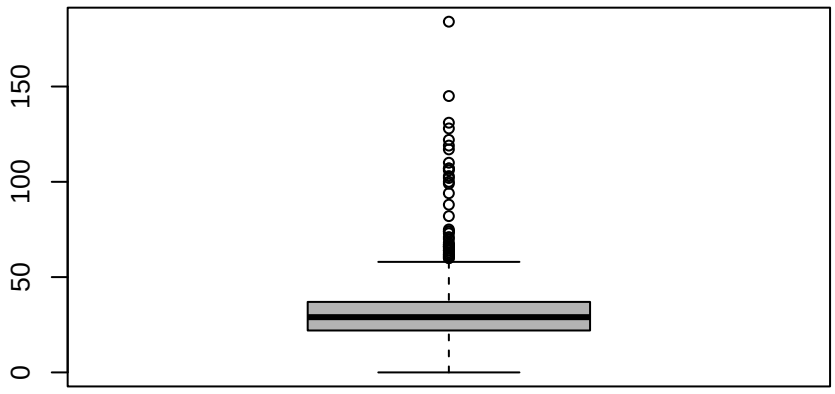
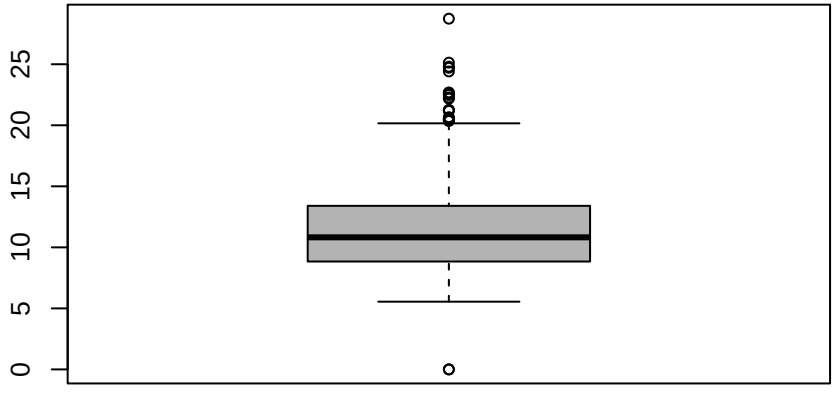
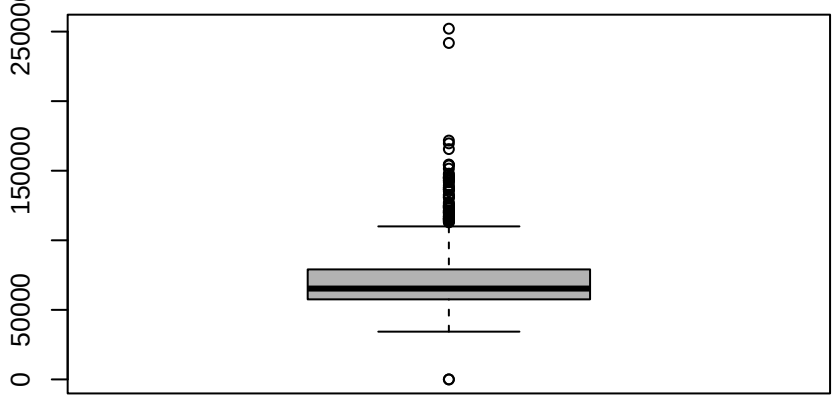
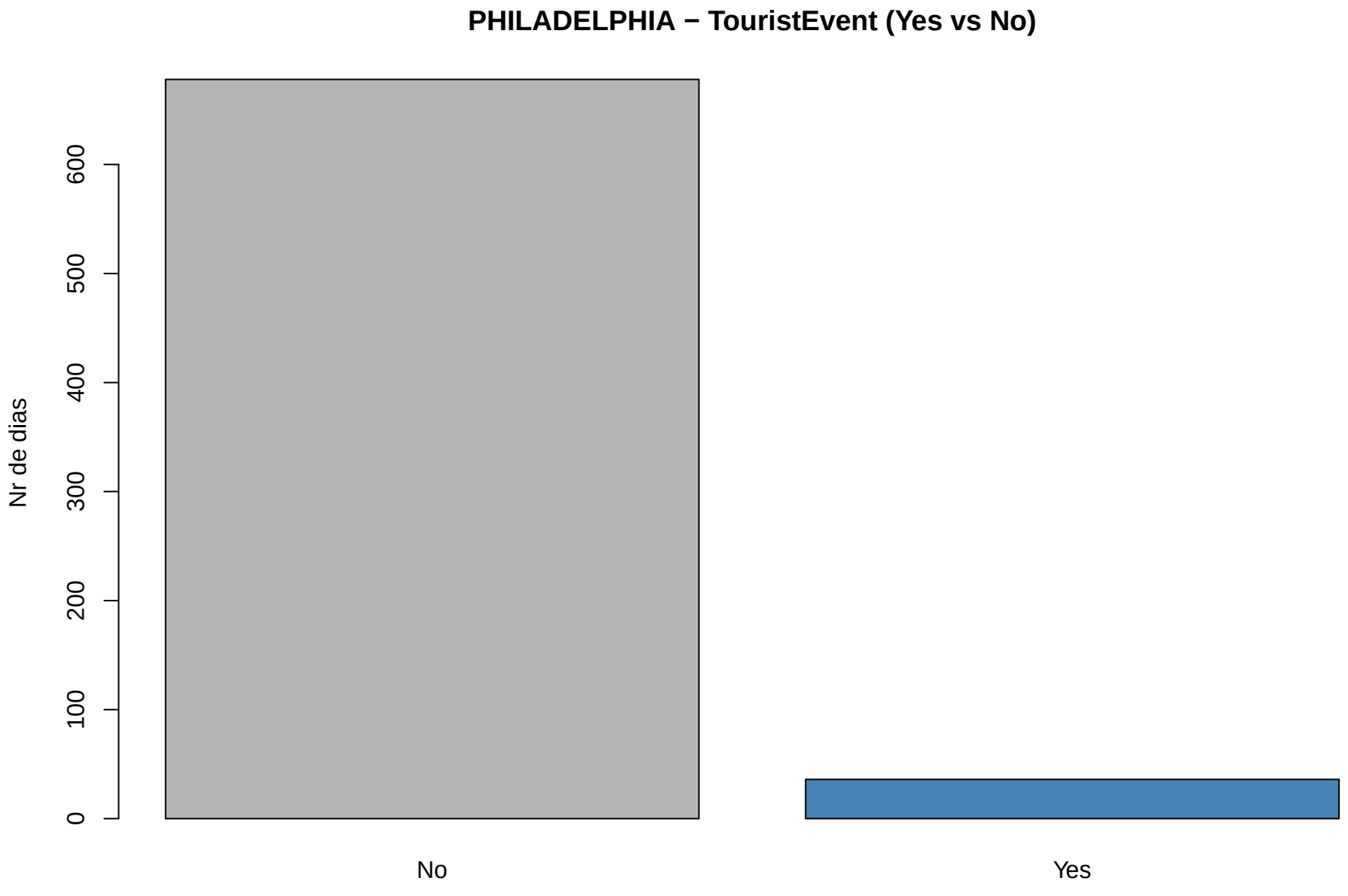

PHILADELPHIA − Num_Customers (TARGET)
Frequency
0
50
100
150
0
500
1000
1500
Clientes
Num_Employees
Frequency
0
20
60
100
140
0
50
100
150
Empregados
Pct_On_Sale
Frequency
0
20
40
60
80
100
0
5
10
15
20
25
30
% Promocao
Sales (endogena)
Frequency
0
50
100
150
200
0
50000
100000
150000
200000
250000
Vendas

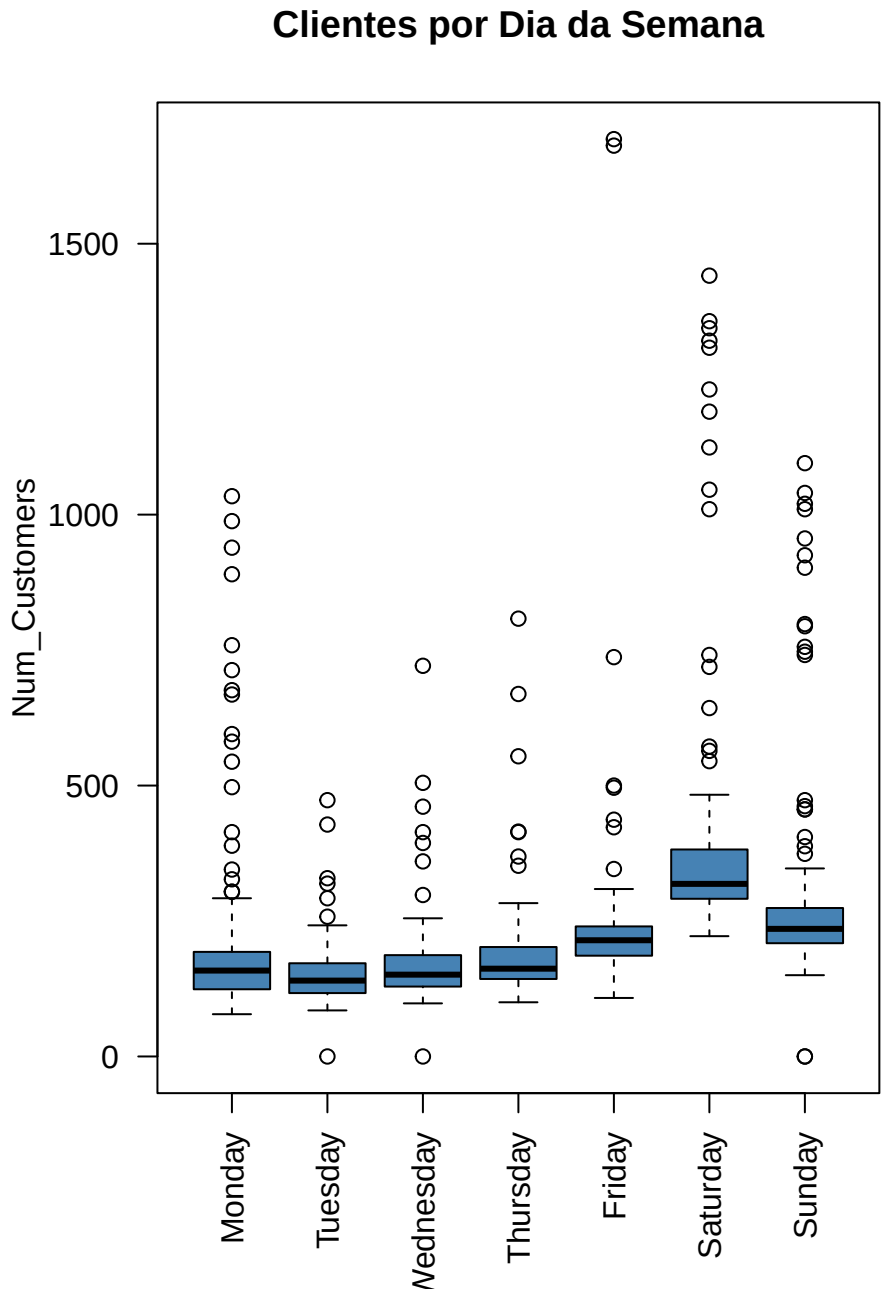

Num_Customers (TARGET)
0
500
1000
1500

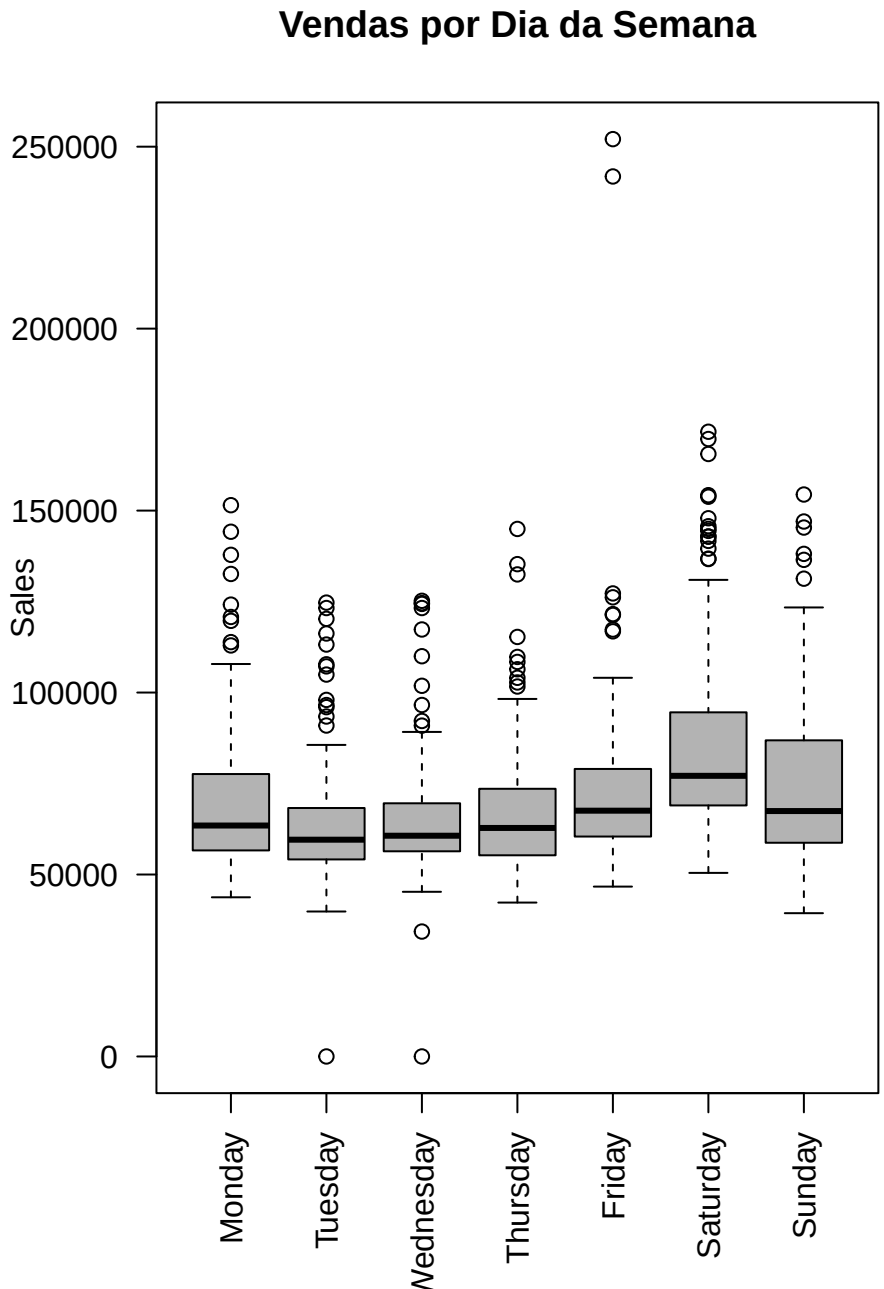

Num_Employees
0
50
100
150

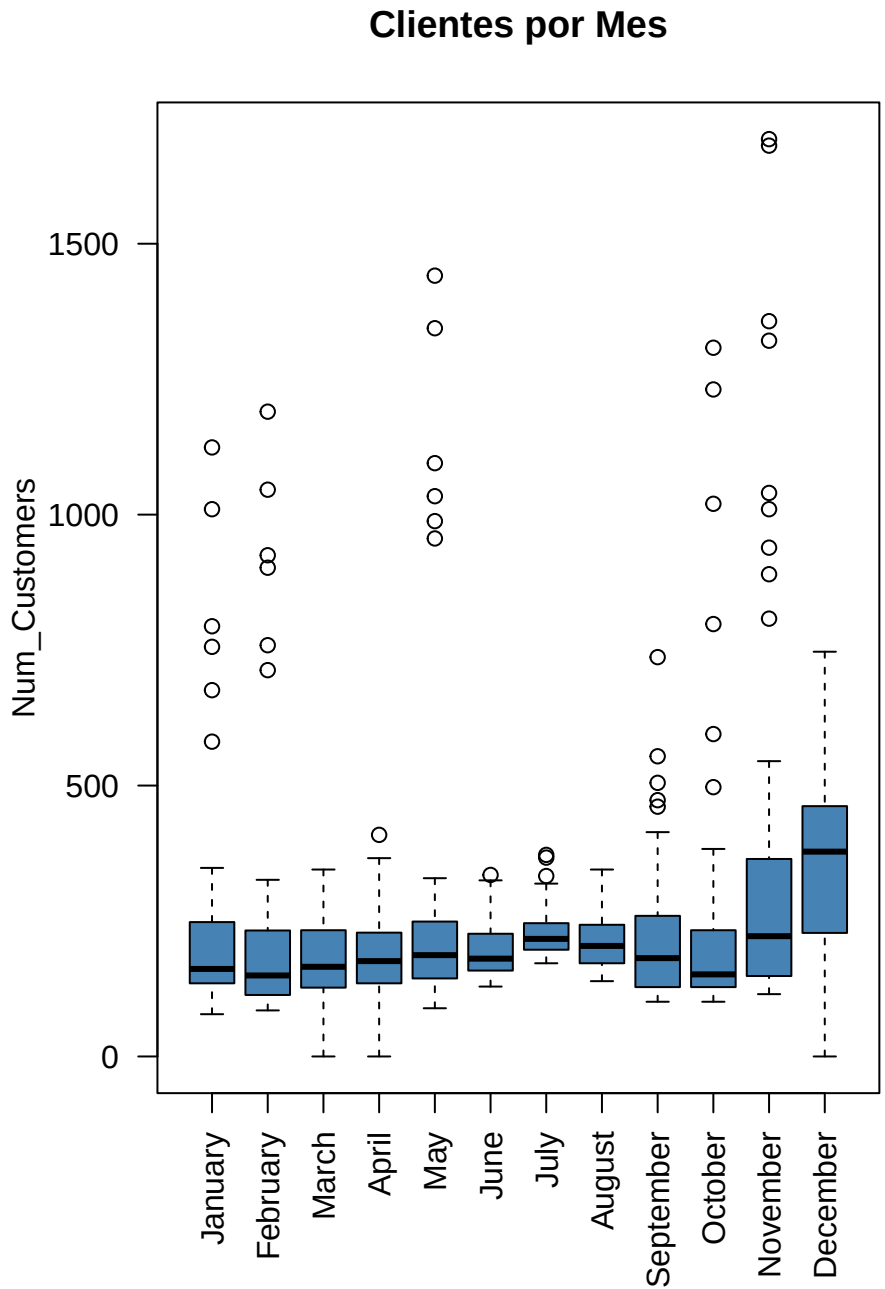

Pct_On_Sale
0
5
10
15
20
25

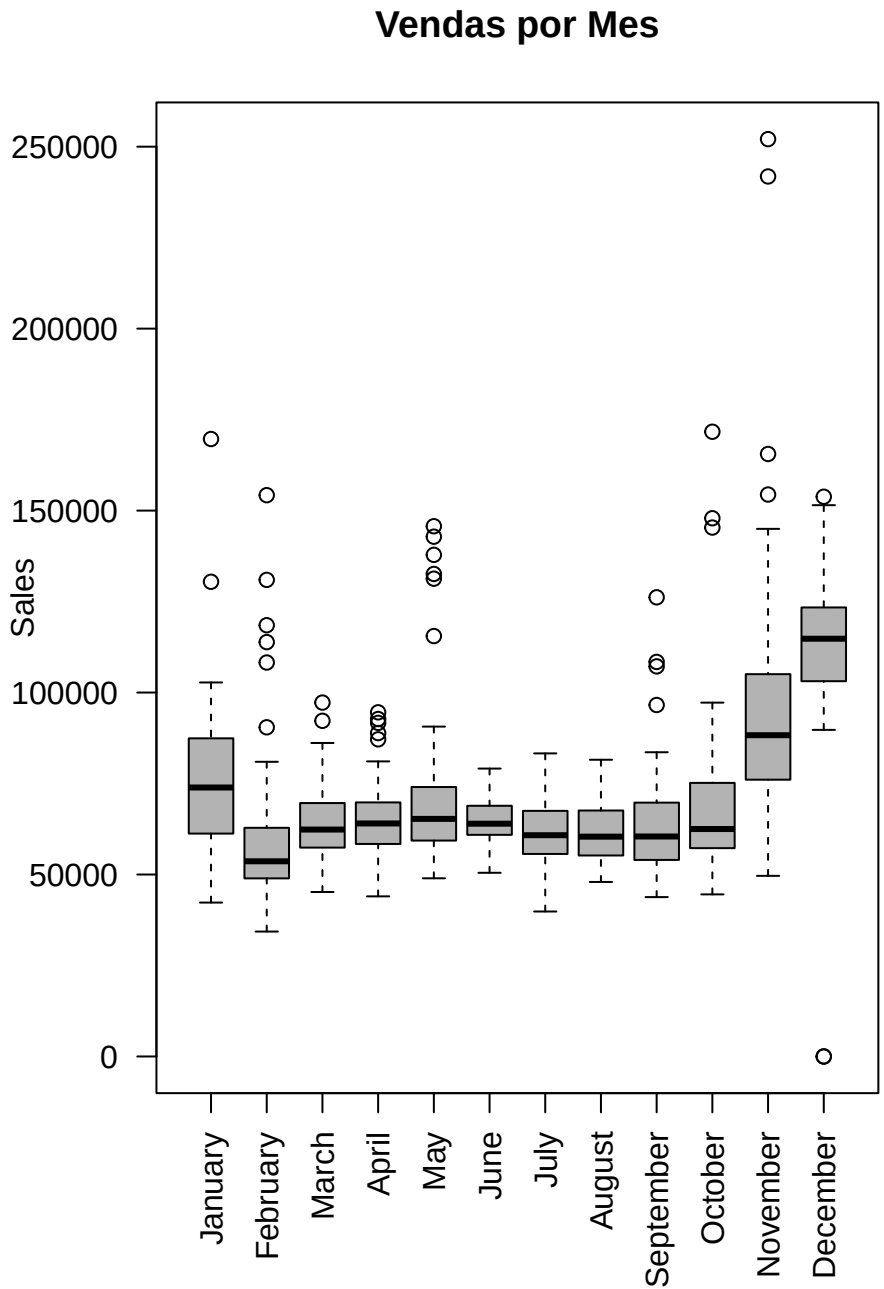

Sales
0
50000
150000
250000


PHILADELPHIA − TouristEvent (Yes vs No)
Nr de dias
0
100
200
300
400
500
600
No
Yes

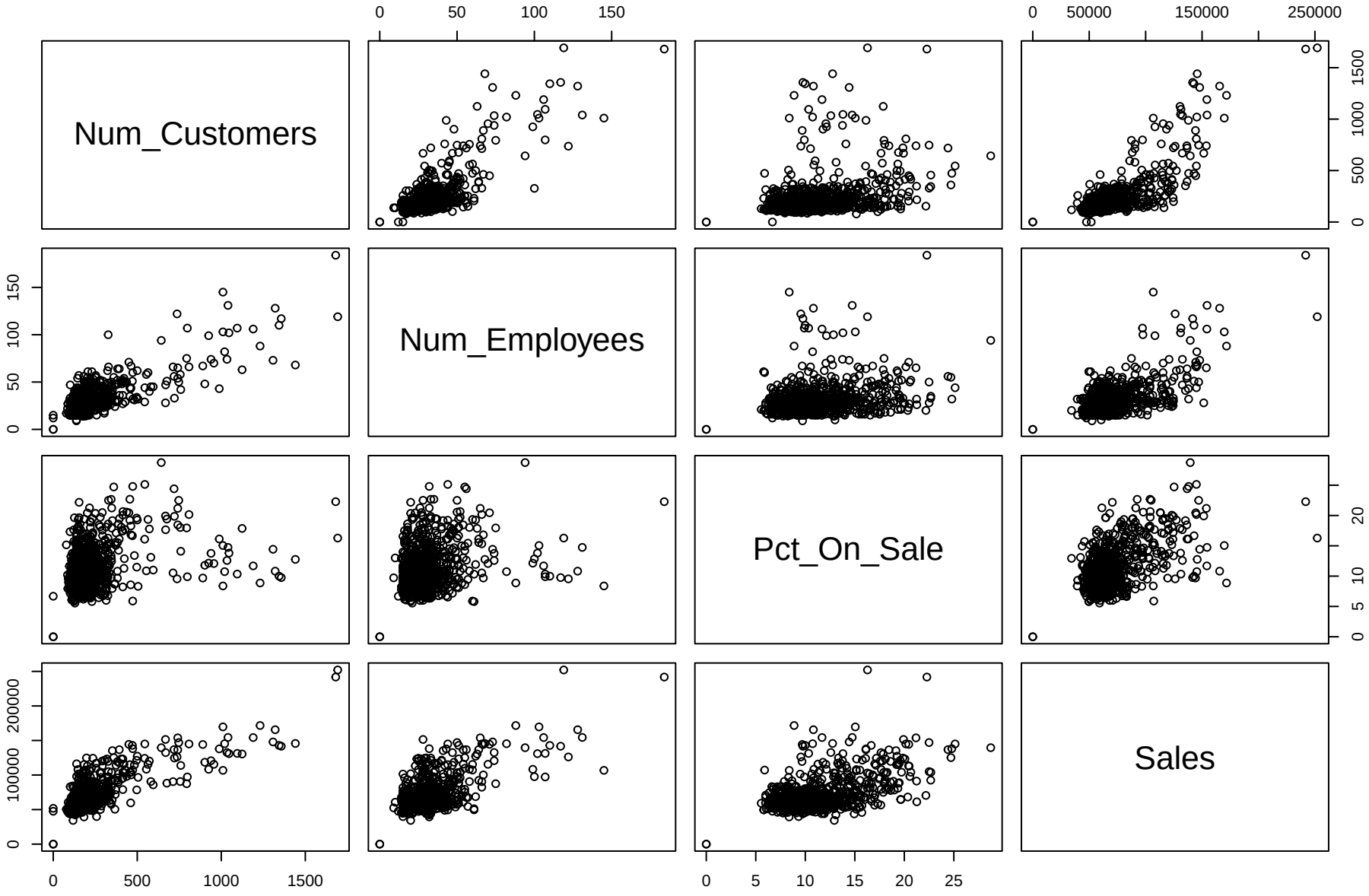

Clientes por Dia da Semana
Num_Customers
0
500
1000
1500
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

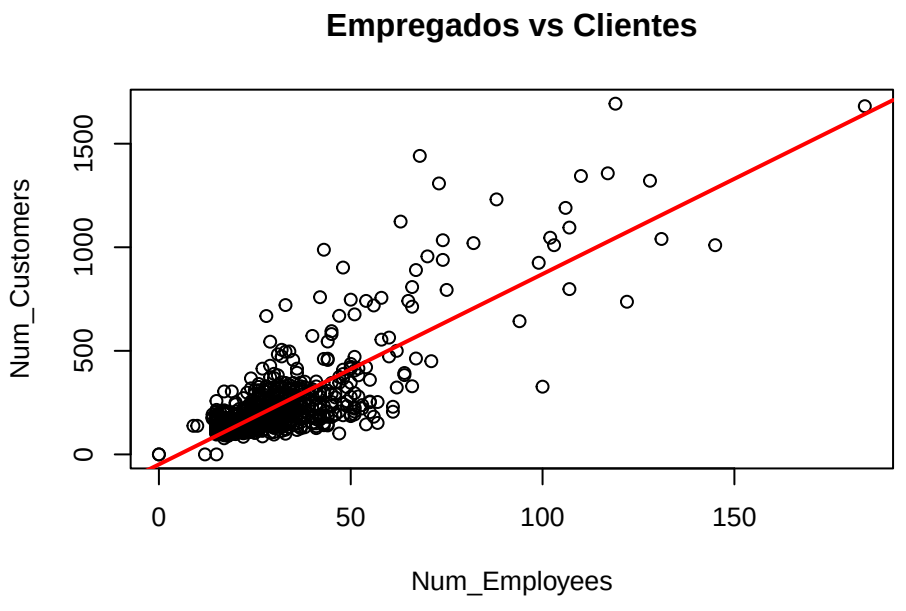

Vendas por Dia da Semana
Sales
0
50000
100000
150000
200000
250000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

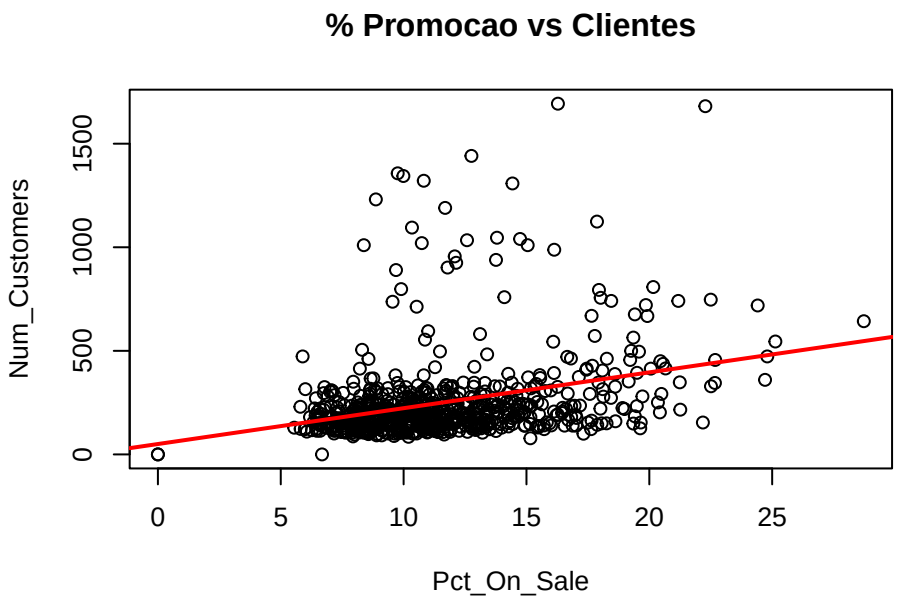

Clientes por Mes
Num_Customers
0
500
1000
1500
January
February
March
April
May
June
July
August
September
October
November
December

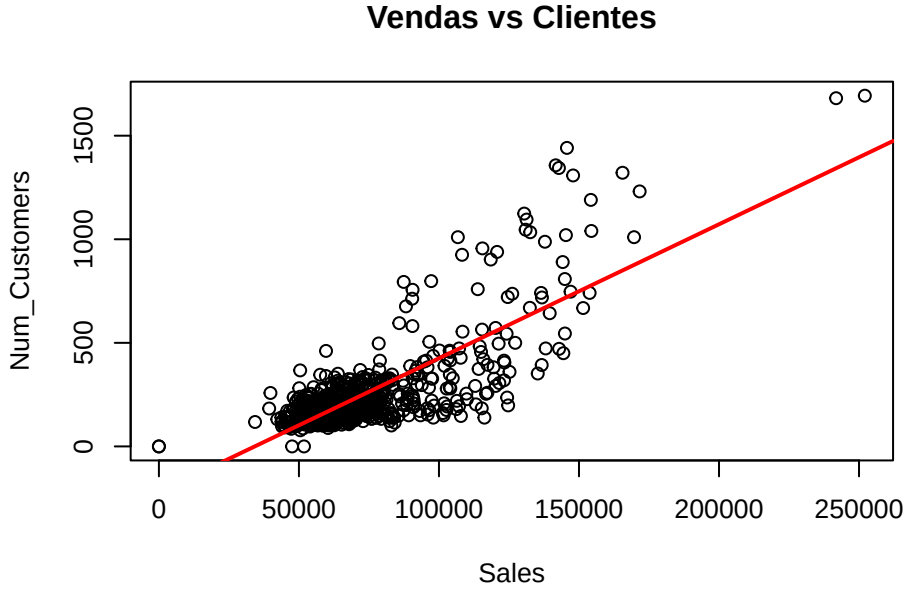

Vendas por Mes
Sales
0
50000
100000
150000
200000
250000
January
February
March
April
May
June
July
August
September
October
November
December

PHILADELPHIA − Scatterplot Matrix

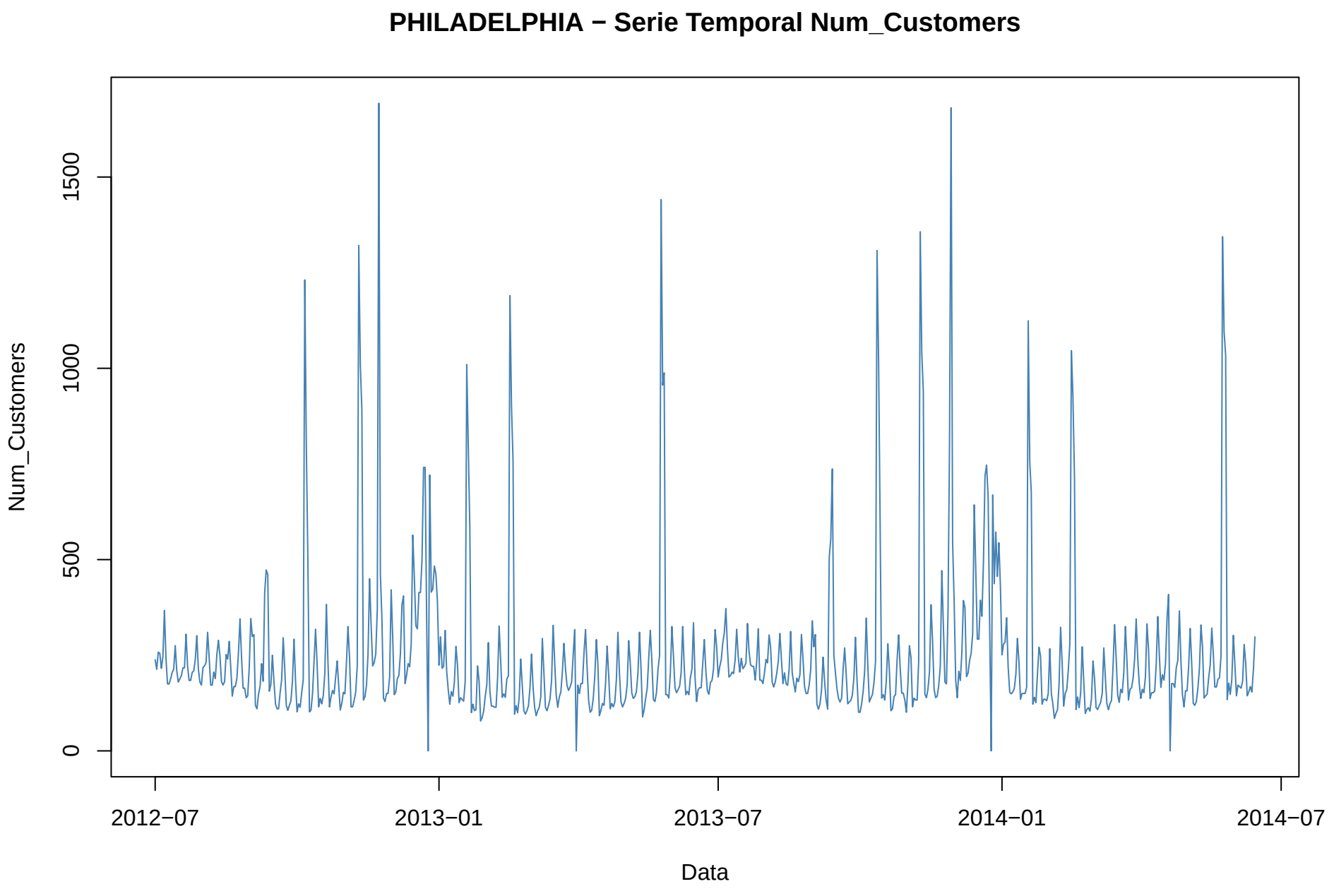


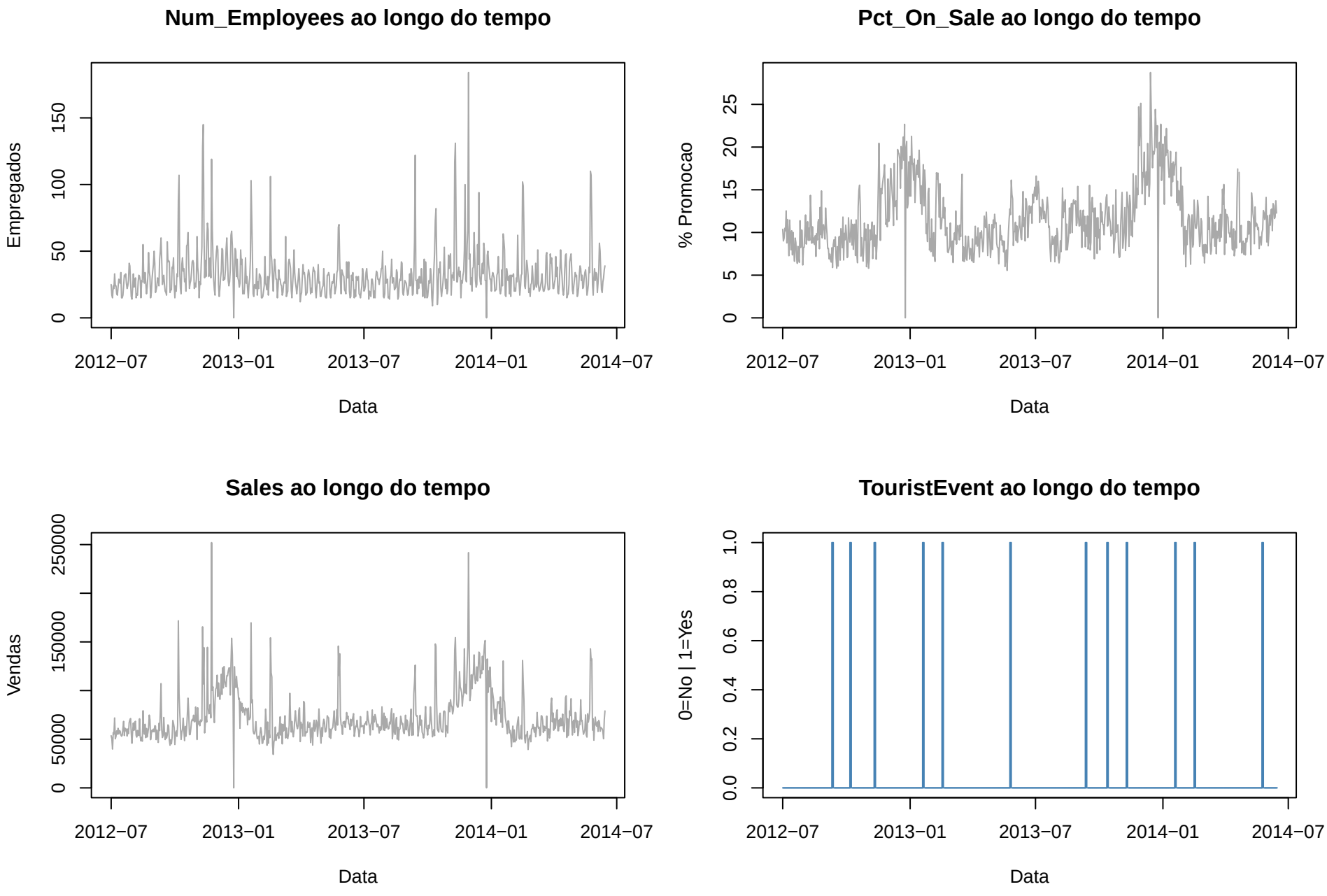

Empregados vs Clientes
Num_Customers
0
500
1000
1500
0
50
100
150
Num_Employees

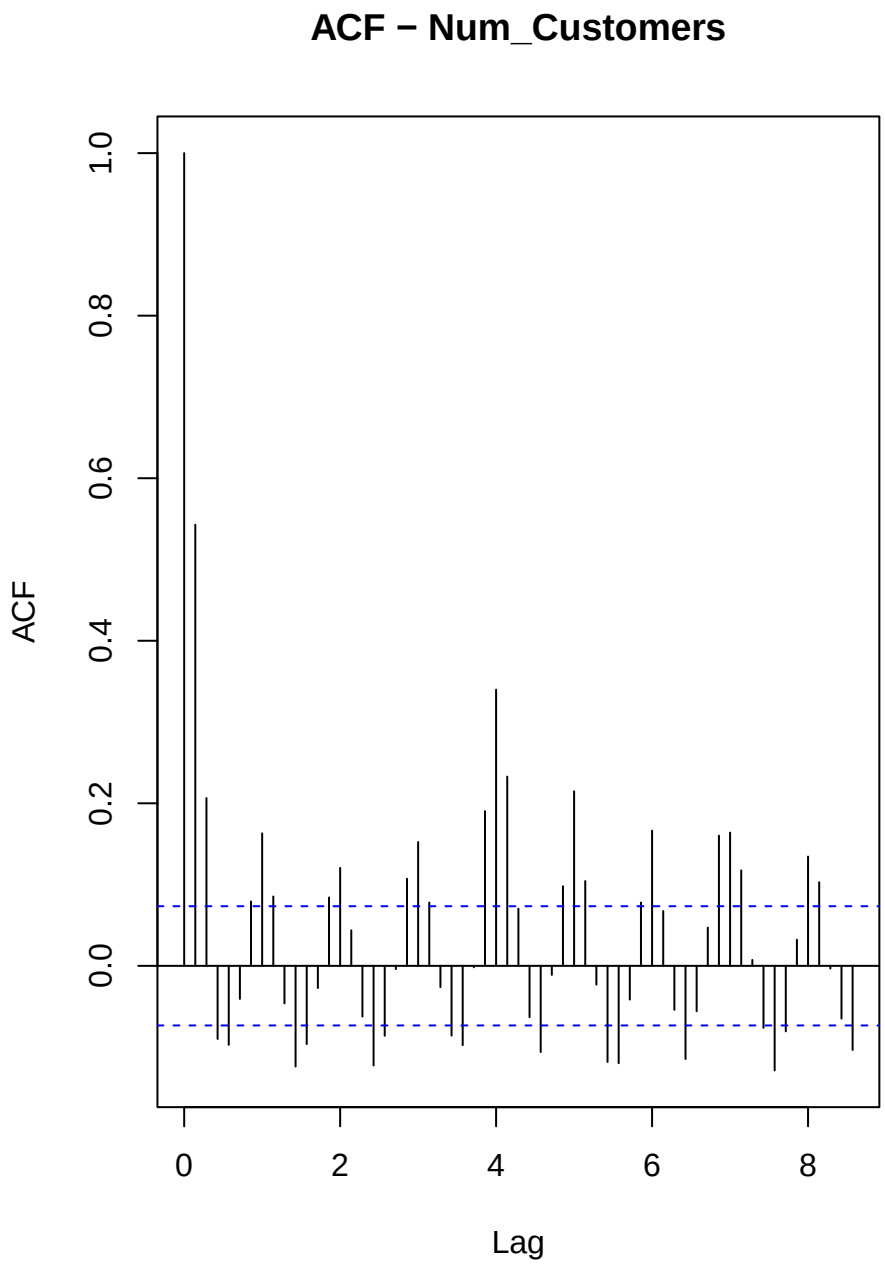

% Promocao vs Clientes
Num_Customers
0
500
1000
1500
0
5
10
15
20
25
Pct_On_Sale

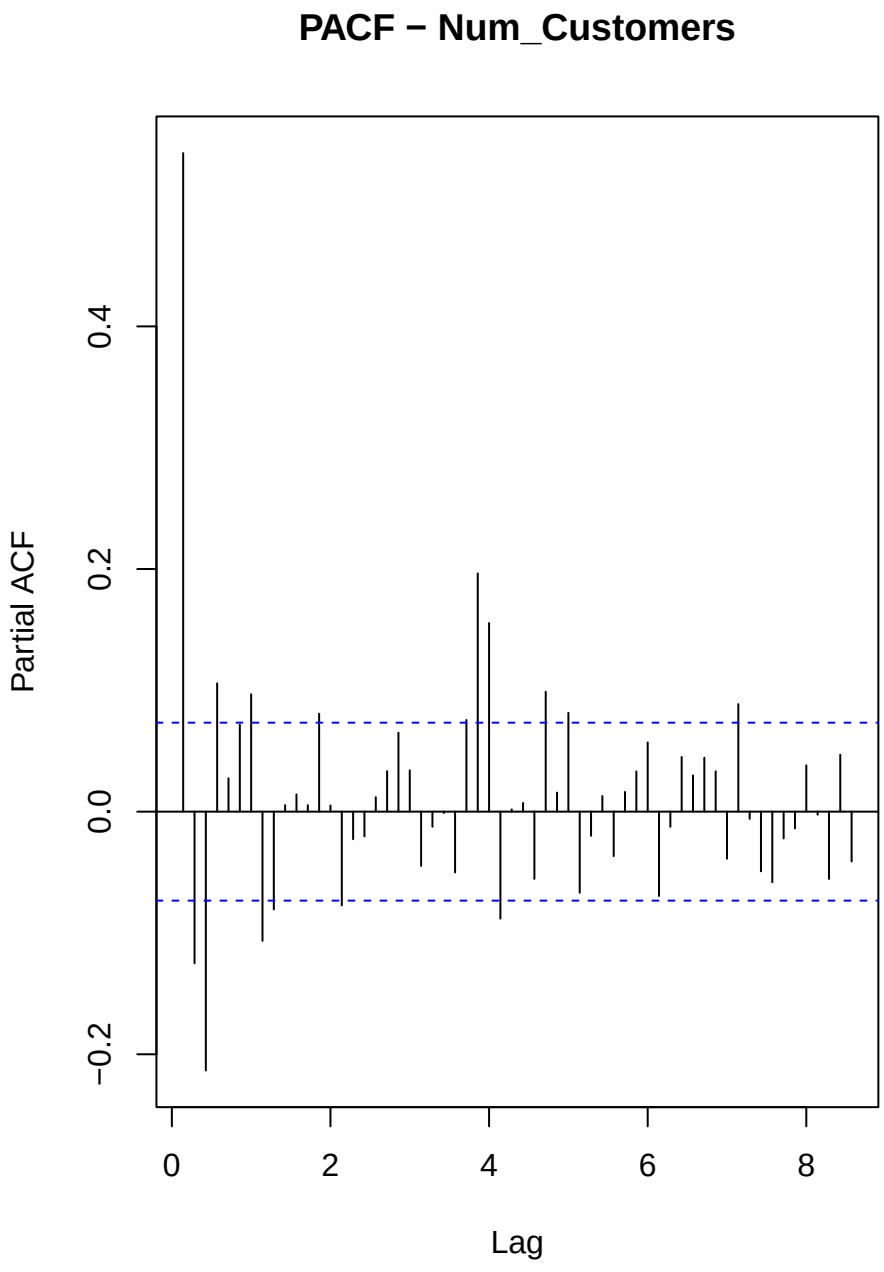

Vendas vs Clientes
Num_Customers
0
500
1000
1500
0
50000
100000
150000
200000
250000
Sales

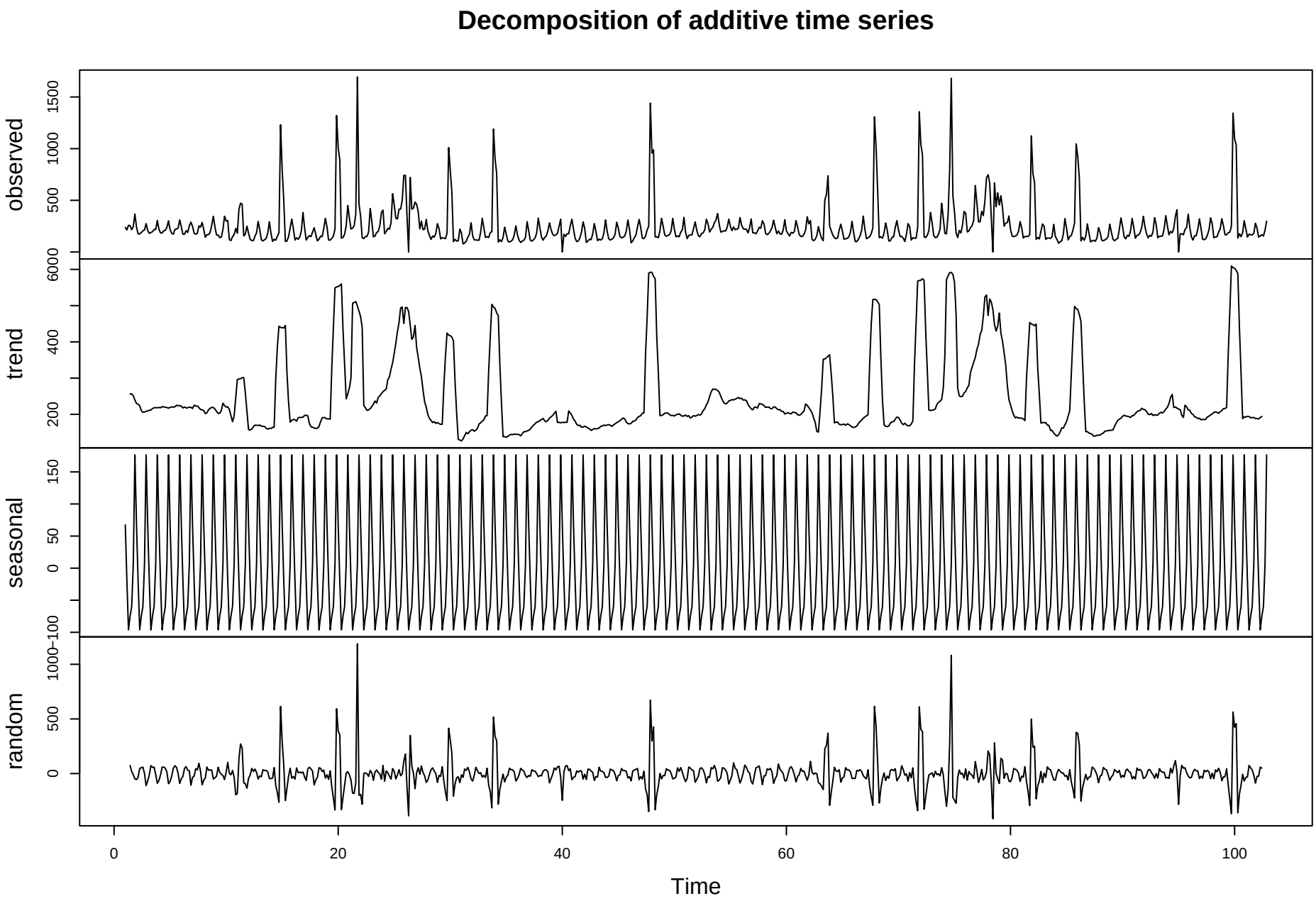
Clientes vs TouristEvent
Num_Customers
0
500
1000
1500
No
Yes
TouristEvent

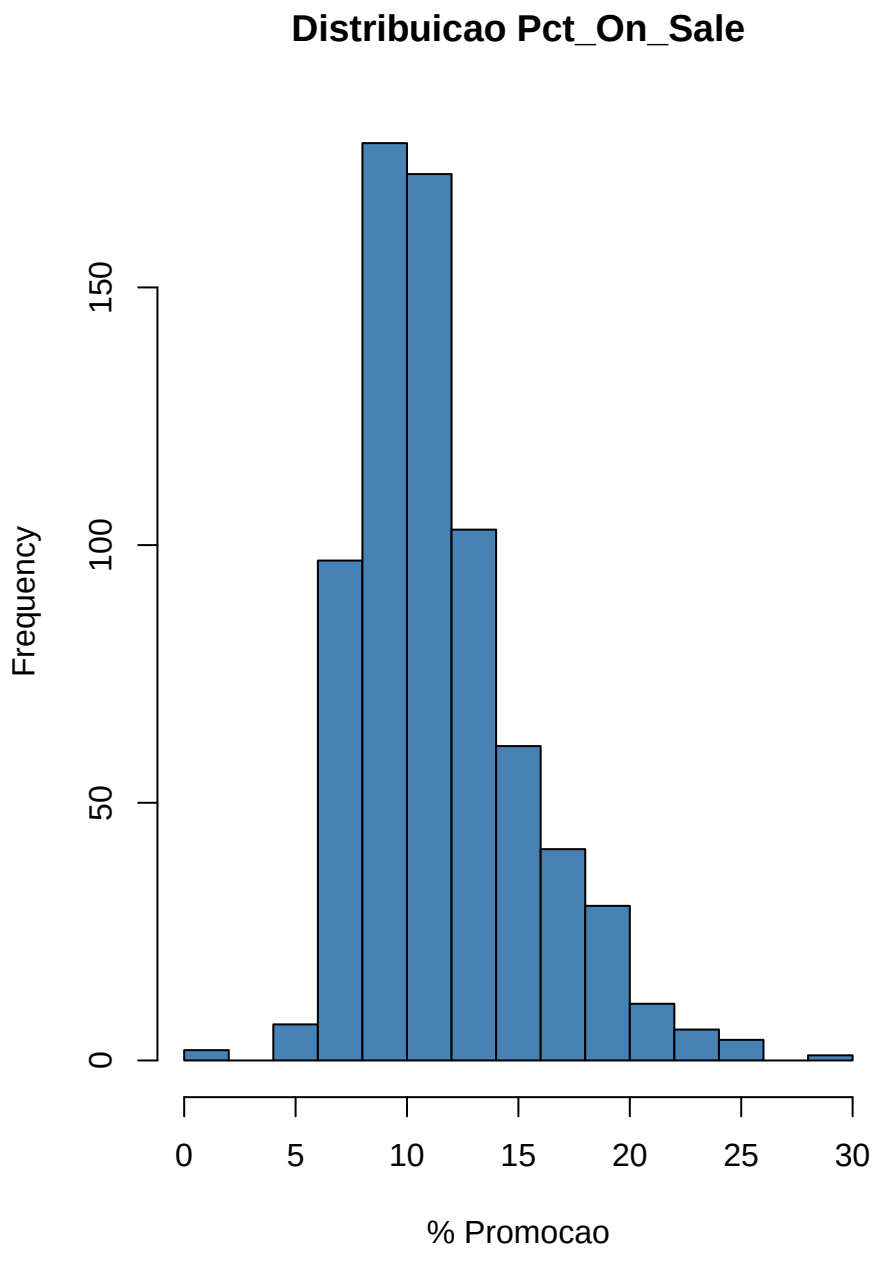

PHILADELPHIA − Serie Temporal Num_Customers
Num_Customers
0
500
1000
1500
2012−07
2013−01
2013−07
2014−01
2014−07
Data

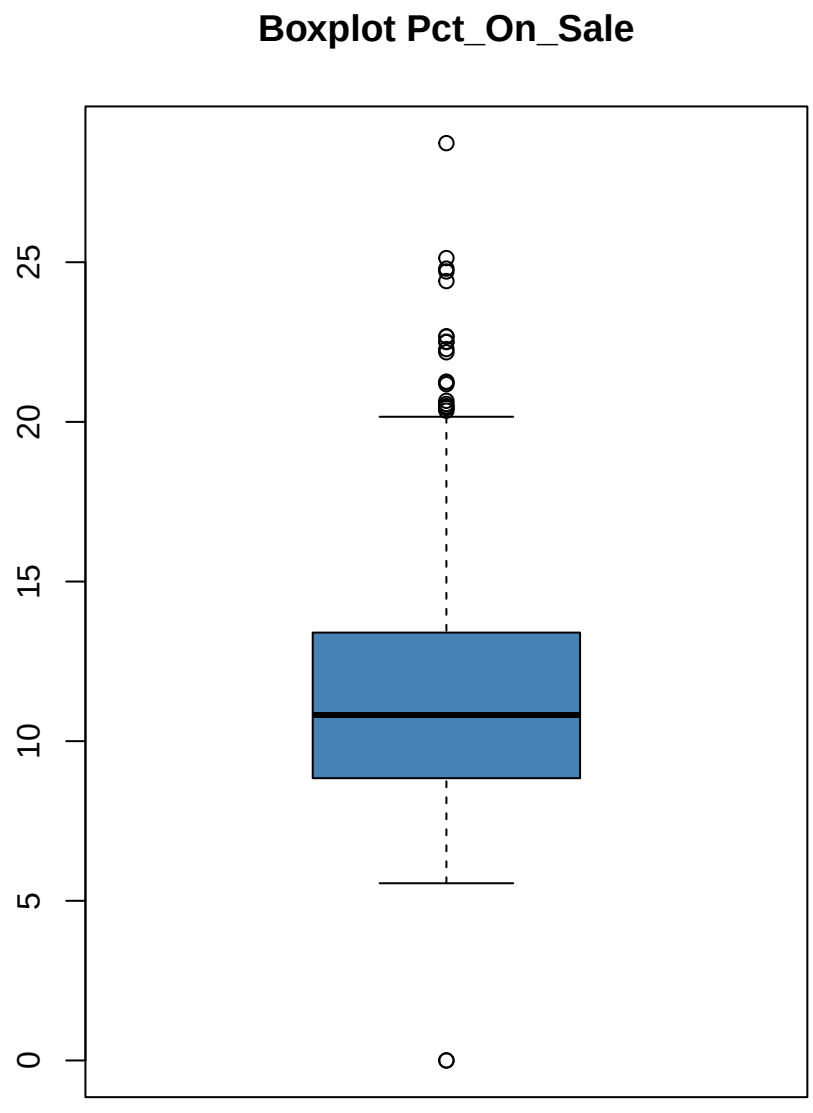

Num_Employees ao longo do tempo
Empregados
0
50
100
150
2012−07
2013−01
2013−07
2014−01
2014−07
Data
Pct_On_Sale ao longo do tempo
% Promocao
0
5
10
15
20
25
2012−07
2013−01
2013−07
2014−01
2014−07
Data
Sales ao longo do tempo
Vendas
0
50000
150000
250000
2012−07
2013−01
2013−07
2014−01
2014−07
Data
TouristEvent ao longo do tempo
0=No | 1=Yes
0.0
0.2
0.4
0.6
0.8
1.0
2012−07
2013−01
2013−07
2014−01
2014−07
Data


ACF − Num_Customers
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Lag


PACF − Num_Customers
Partial ACF
0.4
0.2
0.0
−0.2
0
2
4
6
8
Lag


Decomposition of additive time series
observed
trend
seasonal
random
Time
0
20
40
60
80
100


Distribuicao Pct_On_Sale
Frequency
0
50
100
150
0
5
10
15
20
25
30
% Promocao


Boxplot Pct_On_Sale
0
5
10
15
20
25